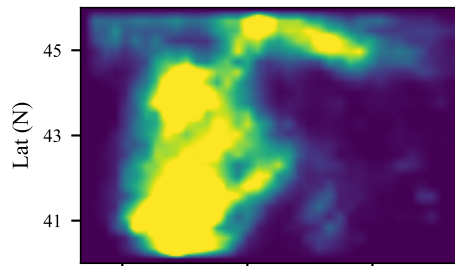
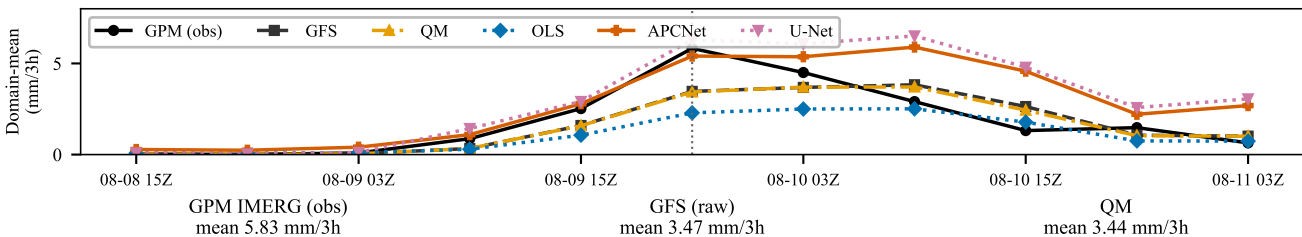


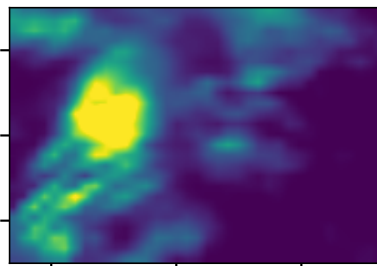
Typical Northeast China cold-vortex heavy-rain event: 2024-08-09 21Z (peak window, domain 25 x 37)

3-h domain-mean precipitation around the peak window



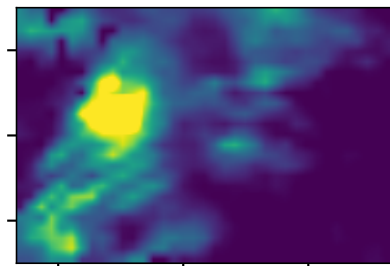
(a)

OLS
mean 2.29 mm/3h



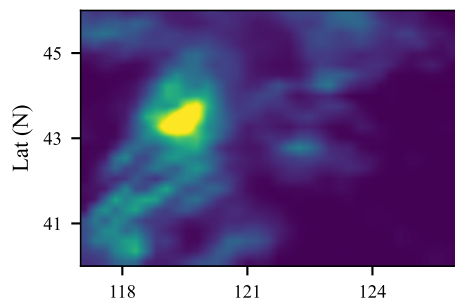
(b)

APCNet
mean 5.40 mm/3h

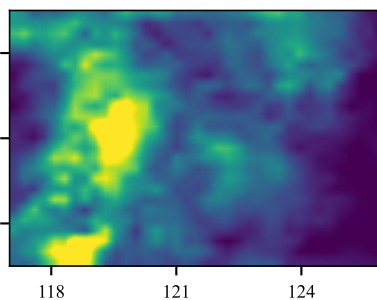


(c)

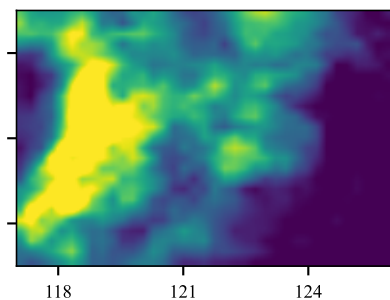
U-Net
mean 6.30 mm/3h



(d)



(e)



(f)

